



SMS Spam Detection System

NLP & Machine Learning — Classifying messages

The Problem

Daily Spam Flood

Fake ads, lottery scams, phishing links — users overwhelmed

Manual Filtering Fails

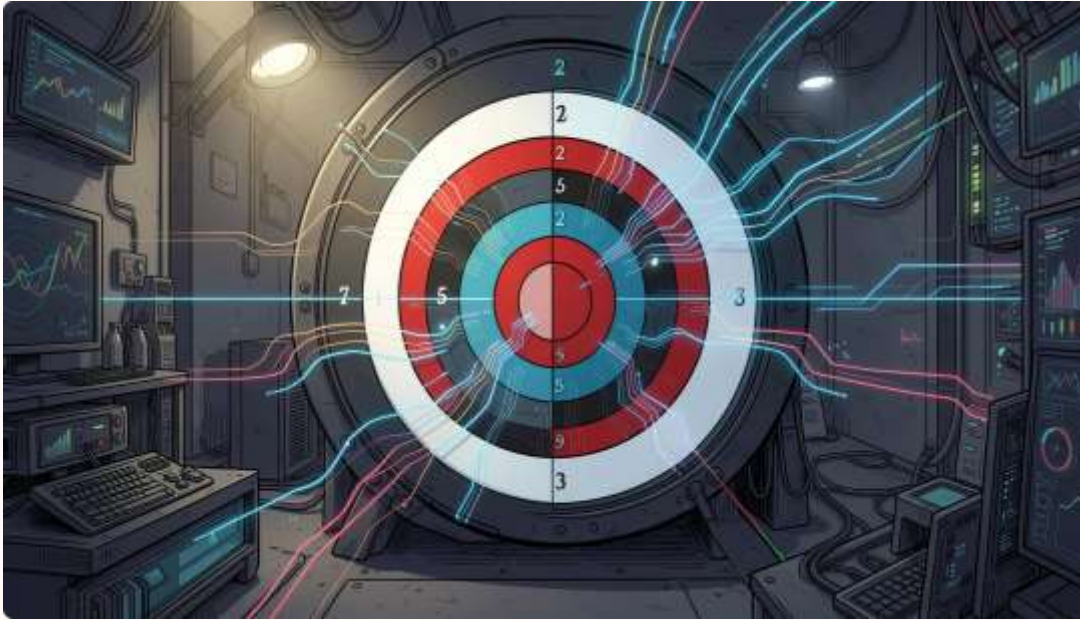
Time-consuming and inefficient at scale

Automated Solution Needed

NLP + ML to classify messages automatically



Project Objectives



-  Preprocess & clean SMS text data
-  Extract features via TF-IDF Vectorization
-  Train ML model for spam classification
-  Evaluate performance with accuracy metrics
-  Predict custom messages as spam or ham

Technologies Used



Python

Core programming language



Scikit-learn

ML library & model training



Multinomial Naive Bayes

Classification algorithm



Pandas & NumPy

Data handling & numerical computation



TF-IDF Vectorizer

Feature extraction from text



Jupyter Notebook

Development environment

Dataset & Preprocessing



MEICA BORTIE-IEAKS.	FIN	ENCIL	BRIVIES	SHOVEL	ENCKOICSOX	FTALA	KAWS
4MWBZ: (75%)			13.1125	11.92820	19.06AD	11	DE 15N
SBAYZ: (359)				\$8,265	490 1.60	718	15M 10M
OAAZY: (564)				\$: 366	\$ 1192	\$,306	171 198
O6AYZ: (202)				\$: 374	1,3898	,760	99K 096
B6ZKZ: (9090)				\$/674	4. 1118	989	GX 105
DAZS: (393)				\$/368	\$,1796	\$1,009	UX 118
DEBYZ: (802)				3.885	\$,2.90	\$,198	681D.346
UGAYZ: (9349)				\$,290	9,406	,306	W 190
O6AYZ: (9099)				\$,388	\$,348	\$1,904	PX 11N
PNIN5S (17795)				\$,598	\$,386	00	FW 10%
AKAX1: (3329)				906	\$,9.96	4179	4X 20.
FI 3875: (3980)					\$,190	\$,4Y	0Y1 10
ABM21 (8990)				\$,806	1,896	880	4Y1 110
MOX27: (8828)				13,966	1,374	\$394	KX
U 5801 (8098)				\$986	2,008	90	NM H5
O 968: (8929)				3835	3,886		4X 10
HA11475: (3380)				34.14			
Q5V171: (8220)							
/SR7Z5: (869)							

SMS Spam Collection Dataset

5,000+ labeled messages — Spam or Ham

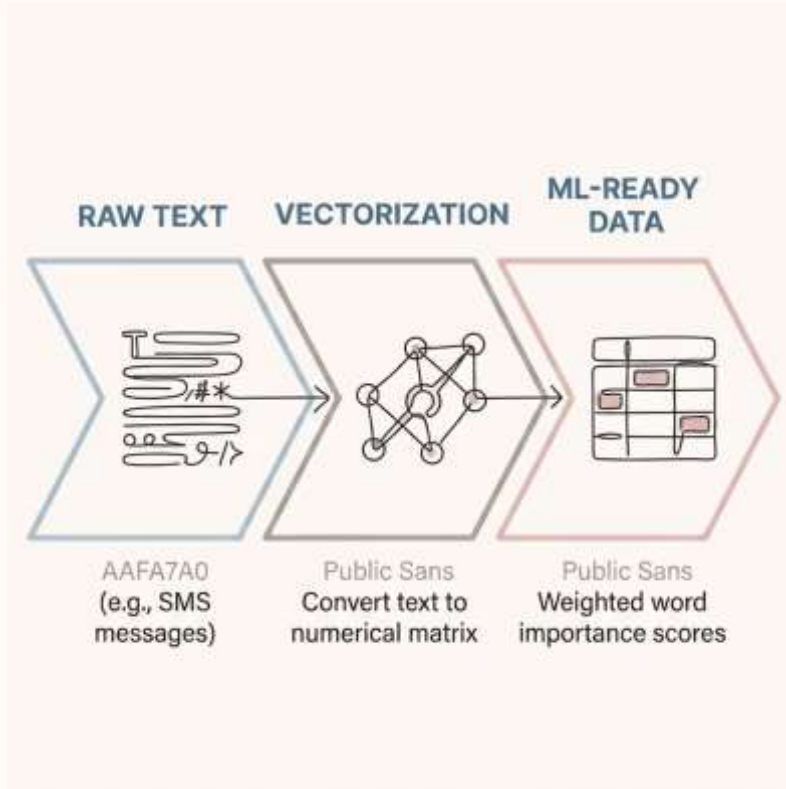
Lowercase Conversion

WIN MONEY → win money

Punctuation Removal

Congratulations!!! →
congratulations

TF-IDF Feature Extraction

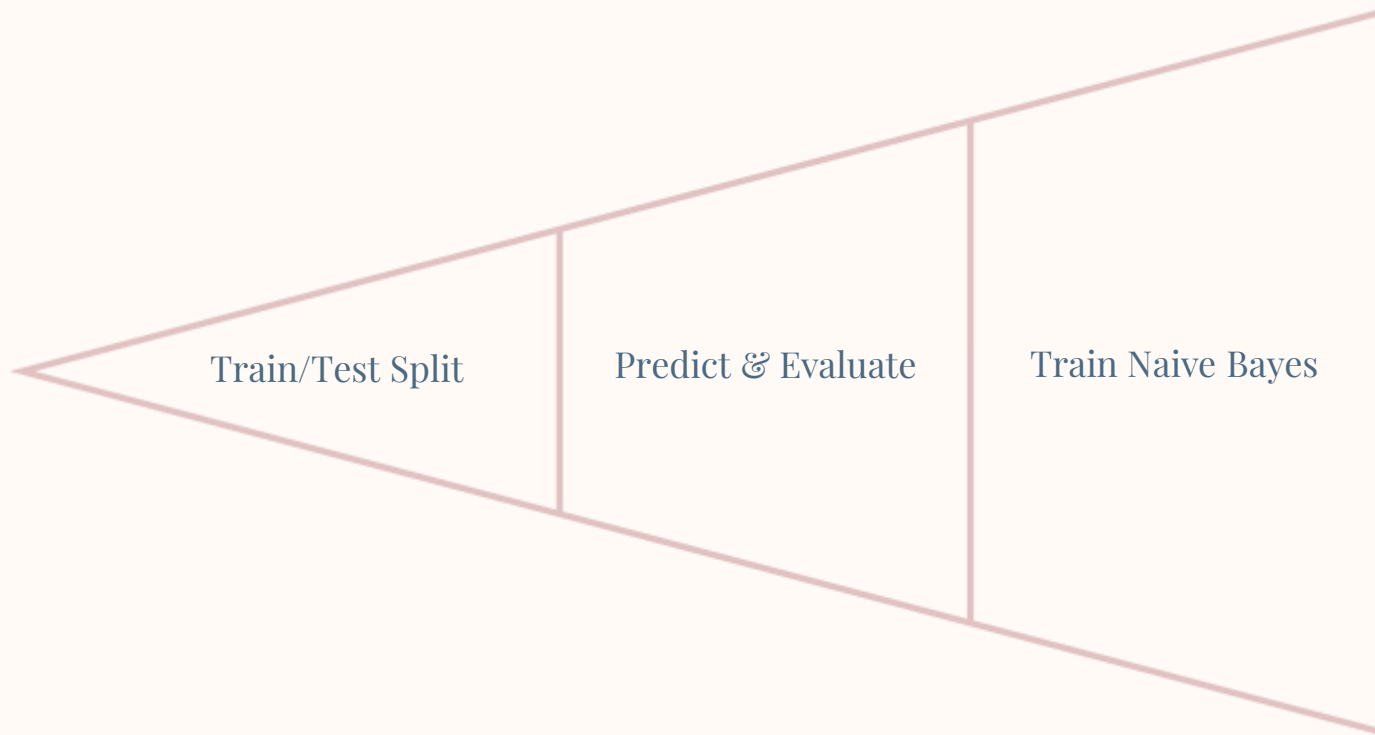


Why TF-IDF?

ML models cannot read text directly. TF-IDF converts messages into numerical vectors, weighting words by importance.

 **Formula:** $TF \times IDF = \text{Word importance score}$

Model Training Pipeline



The Multinomial Naive Bayes classifier is fast, efficient, and well-suited for text classification with TF-IDF vectors.

Model Performance

95–98%

Accuracy Score

Effectively classifies spam vs. ham

5,000+

Messages Analyzed

Labeled dataset for training

80%

Training Data

20% reserved for testing

Advantages & Limitations

✓ Advantages

- Automatic spam detection
- Fast, efficient prediction
- High accuracy (95–98%)
- Beginner-friendly NLP workflow

⚠️ Limitations

- Accuracy depends on dataset quality
- Limited context understanding
- May misclassify complex messages
- English text only

Future Enhancements & Conclusion

Web App

Streamlit or Flask deployment

Deep Learning

LSTM models for better context

Real-Time SMS

Live prediction capability

Word Clouds

Visualize spam word patterns

🟢 **Conclusion:** NLP + Multinomial Naive Bayes achieves **95–98% accuracy** — a strong beginner-level ML & NLP application.

